Backtesting report

Pre requisites before backtesting

Initial capital =1 cr

Capital allocated per trade=2%

Risk free rate(discount rate) = 6%

strategy=
long only,short only, both long and short at same time

instrument=
stocks,stock options,stock futures,index options,index futures,or custom defined stock universe (like nifty 50, nifty 500 or stock universe defined with specific fundamental and technical parameters) commodities etc

timeframe=
15 min,1 hr, daily weekly, monthly and so on

Strategy logic=
purely fundamental,purely technical , purely sentimental,,Multifactor(size,growth,quality,value,momentum) and deep learning algorithms

Backtesting output

Trades log

Month and yearwise return log

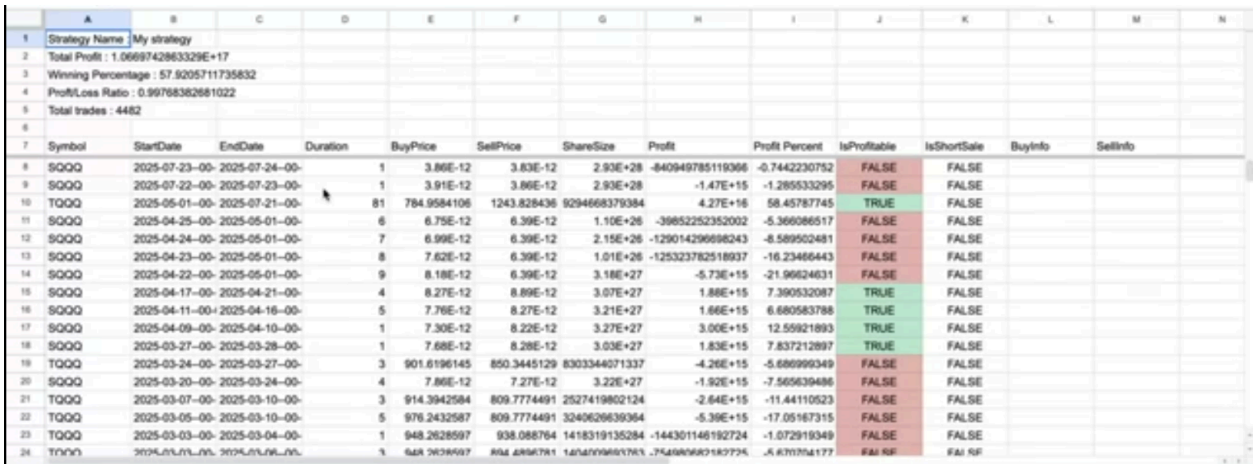
Performance metrics

Equity curve

Drawdown curve

Yearly return histogram

Trades log example



	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	Strategy Name	My strategy												
2	Total Profit : 1.0669742863329E+17													
3	Winning Percentage : 57.9205711735832													
4	Profit/Loss Ratio : 0.99768382681022													
5	Total trades : 4482													
6														
7	Symbol	StartDate	EndDate	Duration	BuyPrice	SellPrice	ShareSize	Profit	Profit Percent	IsProfitable	IsShortSale	BuyInfo	SellInfo	
8	SQQQ	2025-07-23--00-	2025-07-24--00-	1	3.86E-12	3.83E-12	2.93E+28	-840949785119366	-0.7442230752	FALSE	FALSE			
9	SQQQ	2025-07-22--00-	2025-07-23--00-	1	3.91E-12	3.86E-12	2.93E+28	-1.47E+15	-1.285533295	FALSE	FALSE			
10	TQQQ	2025-05-01--00-	2025-07-21--00-	81	784.9584106	1243.828436	92946683793 84	4.27E+16	58.45787745	TRUE	FALSE			
11	SQQQ	2025-04-25--00-	2025-05-01--00-	6	6.75E-12	6.39E-12	1.10E+26	-39852252352002	-5.366086517	FALSE	FALSE			
12	SQQQ	2025-04-24--00-	2025-05-01--00-	7	6.99E-12	6.39E-12	2.15E+26	-129014296698243	-8.589502481	FALSE	FALSE			
13	SQQQ	2025-04-23--00-	2025-05-01--00-	8	7.62E-12	6.39E-12	1.01E+26	-125323782518937	-16.23466443	FALSE	FALSE			
14	SQQQ	2025-04-22--00-	2025-05-01--00-	9	8.18E-12	6.39E-12	3.18E+27	-5.73E+15	-21.96624631	FALSE	FALSE			
15	SQQQ	2025-04-17--00-	2025-04-21--00-	4	8.27E-12	8.89E-12	3.07E+27	1.88E+15	7.390532087	TRUE	FALSE			
16	SQQQ	2025-04-11--00-	2025-04-16--00-	5	7.76E-12	8.27E-12	3.21E+27	1.66E+15	6.680583788	TRUE	FALSE			
17	SQQQ	2025-04-09--00-	2025-04-10--00-	1	7.30E-12	8.22E-12	3.27E+27	3.00E+15	12.55921893	TRUE	FALSE			
18	SQQQ	2025-03-27--00-	2025-03-28--00-	1	7.68E-12	8.28E-12	3.03E+27	1.83E+15	7.837212897	TRUE	FALSE			
19	TQQQ	2025-03-24--00-	2025-03-27--00-	3	901.6196145	850.3445129	8303344071337	-4.26E+15	-5.686999349	FALSE	FALSE			
20	SQQQ	2025-03-20--00-	2025-03-24--00-	4	7.86E-12	7.27E-12	3.22E+27	-1.92E+15	-7.565639486	FALSE	FALSE			
21	TQQQ	2025-03-07--00-	2025-03-10--00-	3	914.3942584	809.7774491	2527419802124	-2.64E+15	-11.44110523	FALSE	FALSE			
22	TQQQ	2025-03-05--00-	2025-03-10--00-	5	976.2432587	809.7774491	3240626639364	-5.39E+15	-17.05167315	FALSE	FALSE			
23	TQQQ	2025-03-03--00-	2025-03-04--00-	1	948.2628597	938.088764	1418319135284	-144301146192724	-1.072919349	FALSE	FALSE			
24	TQQQ	2025-03-01--00-	2025-03-06--00-	3	948.2628597	894.4896781	1418319135284	-764969683192724	-5.670764177	FALSE	FALSE			

NOTE:ADD ONE MORE COLUMN OF CUMULATIVE PROFIT/LOSS IN TRADES LOGBOOK

Month and yearwise return log example



		Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Year
9	1992	22.95	-4.07	6.61	-24.33	7.04	-11.92	-17.18	-4.5	9.63	8.56	18.97	12.56	13.08
10	1993	6.38	-9.19	13.79	-8.77	25.89	-4.3	-15.16	14.72	3.59	3.1	-5.85	8.74	28.58
11	1994	12.01	-2.67	-20.07	3.3	-0.54	-10.7	-2.81	21.9	-3.95	6.99	-6.53	-9.95	-18.07
12	1995	0.33	20.97	10.63	13.73	16.54	36.47	36.05	4.36	-4.3	9.89	-15.33	-7.94	182.65
13	1996	24.56	7.78	9.91	35.14	-1.85	0.27	-7.43	12.38	44.32	7.05	35.39	-3.16	313.45
14	1997	46.29	-19.34	-16.08	18.51	22.63	-16.16	53.96	-3.18	18.53	39.43	7.83	2.81	229.52
15	1998	10.75	29.38	2.39	17.96	-11.91	40.77	-0.05	-31.93	21.48	11.95	-4.42	41.99	169.45
16	1999	80.15	-15.15	23.78	25.99	-24.71	16.84	0.45	9.88	-7.15	-1.14	26.77	34.05	261.01
17	2000	34.77	57.31	13.59	-7.19	-23.75	37.26	-12.14	1.71	-4.74	3.09	21.52	18.82	196.44
18	2001	-14.18	1.44	22.72	-14.95	39.85	25.35	8.32	19.31	4.67	-12.58	21.48	10.41	152.63
19	2002	-8.96	14.87	-22.7	12.59	6.82	16.95	16.84	-17.88	16.7	-17.71	24.92	-5.76	23.34
20	2003	-15.77	-10.94	-7.91	36.9	37.59	-0.32	12.58	10.15	4.48	26.68	0.53	5.71	126.23
21	2004	8.2	-2.03	-7.44	-8.81	14.02	9.88	-16.95	-1.68	2.41	7.34	17.56	9.25	29.21
22	2005	-5.88	-2.23	-4.13	-4.59	11.33	-9.35	14.53	-5.38	3.53	-3.65	18.46	-5.09	3.23
23	2006	11.59	1.41	3.14	-3.34	-0.87	3.66	4.07	6.98	14.2	-2.69	10.02	-5.18	49.62
24	2007	5.62	-7.55	-0.52	16.71	8.03	-2.47	7.68	10.32	3.38	19.69	-15.9	2.62	51.53
25	2008	-4.88	7.93	-8.82	22.79	13.67	12.87	-2.17	-12.36	16.25	24.64	13.03	-11.39	83.48
26	2009	-23.45	6.57	4.23	11.27	24.82	-2.16	29.78	6.41	25.31	-9.78	17.81	20.23	155.53
27	2010	2.62	0.6	-8.59	-7.08	5.75	3	6.52	-3.41	20.08	18.74	-0.44	16.12	62.02
28	2011	11.77	8.78	-3.04	8.26	-5.19	5.88	1.01	1.77	-12.38	17.16	-12	-11.55	5.22
29	2012	17.55	21.18	22	-3.58	-11.49	7.64	2.27	17.29	8.12	10.33	0.19	-5.29	116.78
30	2013	-1.37	-4.84	8.83	5.27	9.81	-5.23	18.24	-1.73	9.95	3.04	2.8	4.07	57.62

Performance metrics in summary

Parameter	Formula/Description	Purpose	Good Value
Total Return	(Ending Capital - Starting Capital) / Starting Capital	Overall profitability	> 20% annual
CAGR	(Ending/Starting)^(1/Years) - 1	Compounded growth rate	> 10%
Sharpe Ratio	(Return - Risk-free rate) / Std Dev	Risk-adjusted return	> 1.0
Sortino Ratio	(Return - Risk-free rate) / Downside Std Dev	Downside risk adjustment	> 2.0
Calmar Ratio	CAGR / Max Drawdown	Return vs worst drawdown	> 0.5
Profit Factor	Gross Profit / Gross Loss	Profit efficiency	> 1.5
Expectancy	(Win% × Avg Win) + (Loss% × Avg Loss)	Average profit per trade	Positive
Win Rate	Winning Trades / Total Trades	Consistency measure	40-60%

Performance metrics in details

Period

Trading Period

Start Date

End Date

Trade Statistics

Total Trades

Win Trades

Loss Trade

Returns

Initial Capital

Final Equity

Total P&L

Realized P&L

Average P&L per Trade

Median P&L

CAGR

Skewness

Risk Metrics

Win Rate

Loss rate

Average Win

Average Loss

Risk/Reward Ratio**(Profit factor)**

Expectancy value(EV)

Sharpe Ratio

Calmar Ratio

Sortino ratio

Streaks

Max Win Streak

Loss Streak

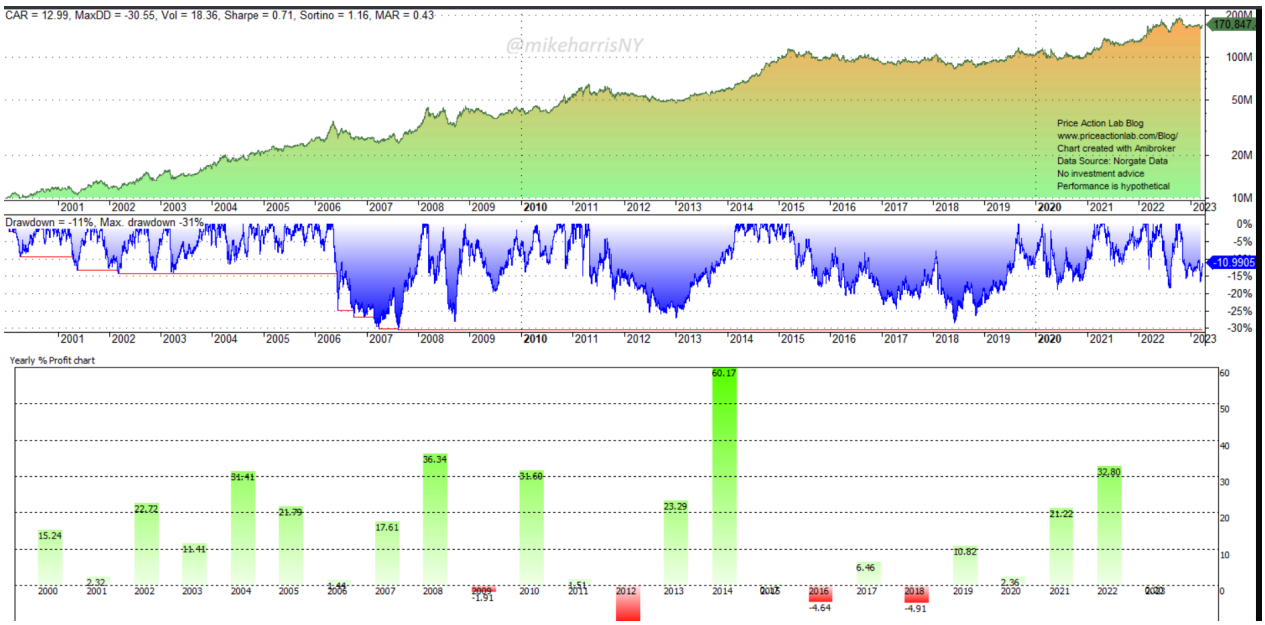
Drawdown

Maximum Drawdown

Max DD Duration

DD Details Peak: Trough:

Equity Curve ,Drawdown curve and Yearly % profit histogram chart example



Skewness

- Positive skew: Long periods of drawdowns, tested patience, but occasional large wins keep traders engaged. Potential for explosive gains but may suffer frequent small losses.
- Negative skew: Regular small wins create overconfidence, leading to complacency about tail risks.Higher probability of extreme losses → higher tail risk.

Calculating Skewness in Trading Data:

Formula:

$$\text{Skewness} = \frac{n}{(n-1)(n-2)} \sum \left(\frac{x_i - \bar{x}}{s} \right)^3$$

Where:

- n = number of trades
- x_i = individual trade P&L
- $\bar{x}$ = mean P&L
- s = standard deviation

Interpretation:

- **Skewness ≈ 0**: Normal/symmetric distribution
- **Skewness > 0**: Positive skew
- **Skewness < 0**: Negative skew

Skewness Analysis of Your Trade Log:

From your provided data, let's calculate the skewness of the `pnl_comm` column:

Summary Statistics:

- **Number of trades:** 4,552
- **Mean P&L:** ₹150.12
- **Median P&L:** ₹-1.42
- **Std Dev:** ₹6,745.82

Skewness Calculation:

Using the formula above:

$$\text{Skewness} \approx \mathbf{18.73}$$

This indicates **extremely positive skewness** in your trading results.

Expectancy value(EV) OR Trading Edge CALCULATION

- Total Trades: 100
- Winning Trades: 60
- Losing Trades: 40
- Total Profit from Winners: ₹60,000
- Total Loss from Losers: ₹20,000

Step 1: Calculate Components

- Win Rate = 60 ÷ 100 = 0.60 (60%)
- Loss Rate = 40 ÷ 100 = 0.40 (40%)
- Average Win = ₹60,000 ÷ 60 = **₹1,000**
- Average Loss = ₹20,000 ÷ 40 = **₹-500** *(negative because it's a loss)*

Step 2: Apply Expectancy Formula

$$\begin{aligned}\text{Expectancy} &= (0.60 \times 1000) + (0.40 \times -500) \\ &= 600 + (-200) \\ &= \mathbf{₹400}\end{aligned}$$